

MERT KAYAALP

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RESEARCH INTERESTS

Machine learning, multi-agent networks, time-series forecasting, causality, financial and statistical signal processing

EXPERIENCE

Researcher with PhD , Dalle Molle Institute for Artificial Intelligence (IDSIA USI-SUPSI) Affiliated with the UBS-IDSIA AI Lab	Sep 2025 - Ongoing Lugano, Switzerland
Postdoctoral Researcher , Adaptive Systems Lab, School of Engineering, EPFL LLM-based social network simulations and cooperative multi-agent systems	Mar 2025 - Aug 2025 Lausanne, Switzerland
Applied Scientist Intern , ML and Forecasting Team, Amazon Web Services Small time-series foundation models and portfolio design	Sep 2024 - Jan 2025 Berlin, Germany
PhD Candidate , School of Computer and Communication Sciences, EPFL Thesis: Sequential Reasoning with Socially Caused Beliefs	Sep 2019 - Feb 2025 Lausanne, Switzerland
Research Internship , Tandon School of Engineering, New York University Communication and Information Theory (CommIT) Lab: Fundamental limits of caching	Jun 2018 - Sep 2018 New York, US
Software Engineer Intern , Comodo Cybersecurity Contribution to database migration project with Java and MongoDB	Jun 2017 - Jul 2017 Ankara, Turkey

EDUCATION

Ph.D., École Polytechnique Fédérale de Lausanne (EPFL) Computer and Communication Sciences Thesis: Sequential Reasoning with Socially Caused Beliefs	Sep 2019 - Feb 2025 Lausanne, Switzerland Advisor: Prof. Ali H. Sayed
B.S., Koç University Electrical and Electronics Engineering, Summa Cum Laude	Sep 2015 - Jun 2019 Istanbul, Turkey
Ankara High School of Science International Physics Olympiads (IPhO) – Bronze Medalist	Sep 2011 - Jul 2015 Ankara, Turkey

HONORS AND AWARDS

Best Student Paper Award – IEEE Statistical Signal Processing Workshop	Hanoi, Vietnam, Jul 2023
Rising Stars in Signal Processing - IEEE ICASSP	Rhodes, Greece, Jun 2023
EPFL EDIC Fellowship - For the first year of PhD studies	Sep 2019 - Sep 2020
Vehbi Koç Honor Awardee - Awarded to students achieving the highest GPAs	Feb 2016 - Jun 2019
Semahat Arsel Scholar - Full tuition scholarship with stipend for top performers in LYS	Sep 2015 - Jun 2019
Bronze Medal - International Physics Olympiad (IPhO)	Mumbai, India, Jul 2015
Silver Medal - National Physics Olympiad (Turkish National Science Foundation)	Dec 2014, Dec 2013
Turkish National University Entrance Examination (LYS) 1st out of 1.8 million participants	Jun 2015

SKILLS

Programming • Python, Java, MATLAB • PyTorch, TensorFlow, Scikit-learn, Pandas • Amazon SageMaker, Bedrock, AutoGluon	Mathematics • Probability, Stochastic Processes, Optimization • Social Networks, Graphical Models • Reinforcement Learning, Causal Inference
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RECENT TALKS

1. “Causal Influences in Interacting Networks”, *Research seminar at IDSIA*, Lugano, October 2025
2. “Modeling Decision-Making over Decentralized Networks: Information Fusion and Influence”, *Amazon Tech Talk*, Barcelona, June 2025
3. “Decision Making in an Interconnected and Modular World”, *Princeton Workshop on Optimization, Learning, and Control*, Princeton, New Jersey, June 2024
4. “Mathematical Models of Social Opinion Formation”, *Invited talk at EPFL-Polympiads*, Lausanne, April 2024
5. “Collaborative Decision Making over Ad-hoc Networks”, *Invited talk at Ford Otosan Automotive Technologies Laboratory*, Online, December 2023
6. “Interpretable and Socially Intelligent Systems”, *Rising Stars in Signal Processing Workshop, IEEE ICASSP*, Rhodes, June 2023

PUBLICATIONS

JOURNAL

1. M. Kayaalp and A. H. Sayed, "Causal influences over social learning networks," *Journal of Machine Learning Research*, May 2026
2. P. Hu, M. Kayaalp, and A. H. Sayed, "Minimizing the probability of error for decision making over graphs," *IEEE Open Journal of Signal Processing*, vol. 6, pp. 1139-1160, October 2025
3. P. Hu, V. Bordignon, M. Kayaalp, and A. H. Sayed, "Non-asymptotic performance of social machine learning under limited data," *Signal Processing*, p. 109849, December 2024
4. M. Kayaalp, Y. Inan, V. Koivunen, and A. H. Sayed, "Causal influence in federated edge inference," *IEEE Transactions on Signal Processing*, vol. 72, pp. 5604-5615, November 2024
5. V. Shumovskaia, M. Kayaalp, and A. H. Sayed, "Detection of malicious agents in social learning," *IEEE Signal Processing Letters*, vol. 31, pp. 1745-1749, July 2024
6. V. Shumovskaia, M. Kayaalp, and A. H. Sayed, "Social learning in community structured graphs," *IEEE Transactions on Signal Processing*, vol. 72, pp. 2812-2826, May 2024
7. M. Kayaalp, Y. Inan, E. Telatar, and A. H. Sayed, "On the arithmetic and geometric fusion of beliefs for distributed inference," *IEEE Transactions on Automatic Control*, vol. 69, no. 4, pp. 2265-2280, April 2024
8. M. Kayaalp, V. Bordignon, and A. H. Sayed, "Social opinion formation and decision making under communication trends," *IEEE Transactions on Signal Processing*, vol. 72, pp. 506-520, January 2024
9. M. Kayaalp, F. Ghadieh, and A. H. Sayed, "Policy evaluation in decentralized POMDPs with belief sharing," *IEEE Open Journal of Control Systems*, vol. 2, pp. 125-145, June 2023
10. V. Shumovskaia, M. Kayaalp, M. Cemri, and A. H. Sayed, "Discovering influencers in opinion formation over social graphs," *IEEE Open Journal of Signal Processing*, vol. 4, pp. 188-207, March 2023
11. M. Kayaalp, S. Vlaski, and A. H. Sayed, "Dif-MAML: Decentralized multi-agent meta-learning," *IEEE Open Journal of Signal Processing*, vol. 3, pp. 71-93, January 2022

CONFERENCE

1. I. Yazici, M. Kayaalp, S. Taga, and A. H. Sayed, "Opinion consensus formation among networked large language models," *IEEE ICASSP*, Barcelona, Spain, May 2026
2. Z. Firsova, M. Khammassi, M. Kayaalp, and A. H. Sayed, "Non-Bayesian social learning for modeling interacting large language model agents," *IEEE ICASSP*, Barcelona, Spain, May 2026
3. M. Kayaalp, C. Turkmen, O. Shchur, P. Mercado, A. F. Ansari, M. Bohlke-Schneider, and B. Wang, "Test-time efficient pretrained model portfolios for time series forecasting," *International Conference on Learning Representations (ICLR)*, Rio de Janeiro, Brazil, April 2026
4. P. Hu, M. Kayaalp, and A. H. Sayed, "Test-time collective classification over multi-agent networks," *NeurIPS 2025 Workshop on Algorithmic Collective Action*, San Diego, US, December 2025
5. P. Hu, M. Kayaalp, and A. H. Sayed, "Optimal combination policies for error exponent maximization in social learning," *IEEE ICASSP*, Hyderabad, India, April 2025
6. M. Kayaalp, Y. Inan, V. Koivunen, and A. H. Sayed, "Causal impact analysis for asynchronous decision making," *IEEE ISIT*, Athens, Greece, 2024
7. V. Shumovskaia, M. Kayaalp, and A. H. Sayed, "Distributed decision-making for community structured networks," *IEEE ICASSP*, Seoul, Korea, April 2024
8. V. Bordignon, M. Kayaalp, V. Matta, and A. H. Sayed, "Social learning with non-Bayesian local updates," *EUSIPCO*, Helsinki, Finland, September 2023
9. M. Kayaalp, Y. Inan, V. Koivunen, E. Telatar, and A. H. Sayed, "On the fusion strategies for federated decision making," *IEEE Statistical Signal Processing Workshop (SSP)*, Hanoi, Vietnam, July 2023, (**award winning article**).
10. V. Shumovskaia, M. Kayaalp, and A. H. Sayed, "Identifying opinion influencers over social networks," *IEEE ICASSP*, Rhodes Island, Greece, June 2023
11. P. Hu, V. Bordignon, M. Kayaalp, and A. H. Sayed, "Performance of social machine learning under limited data," *IEEE ICASSP*, Rhodes Island, Greece, June 2023
12. M. Cemri, V. Bordignon, M. Kayaalp, V. Shumovskaia, and A. H. Sayed, "Asynchronous social learning," *IEEE ICASSP*, Rhodes Island, Greece, June 2023
13. Y. Inan, M. Kayaalp, E. Telatar, and A. H. Sayed, "Social learning under randomized collaborations," *IEEE ISIT*, Helsinki, Finland, June 2022
14. M. Kayaalp, V. Bordignon, S. Vlaski, and A. H. Sayed, "Hidden Markov modeling over graphs," *IEEE Data Science and Learning Workshop (DSLW)*, Singapore, May 2022
15. Y. Inan, M. Kayaalp, A. H. Sayed, and E. Telatar, "A fundamental limit of distributed hypothesis testing under memoryless quantization," *IEEE ICC*, Seoul, South Korea, May 2022
16. M. Kayaalp, S. Vlaski, and A. H. Sayed, "Distributed meta-learning with networked agents," *EUSIPCO*, Dublin, Ireland, August 2021

PATENTS

1. M. Kayaalp and A. H. Sayed, "Automatic identification of causal influences and influential agents in social networks", U.S. patent application, filed June 2024

THESIS

1. M. Kayaalp, "Sequential Reasoning with Socially Caused Beliefs", *École Polytechnique Fédérale de Lausanne*, February 2025

PRE-PRINTS

1. M. Kayaalp and O. Szehr, "Signal recovery from time and frequency samples," *arXiv:2603.16242*, March 2026